

What are Attributes?

Attributes (also called **fields**) are variables within a class. They represent the data or state of an object.

```
public class Main {  
    int x = 5; // Attribute  
    String fname = "John"; // Attribute  
}
```

Accessing Attributes

Use the **dot (.) initialization** to access or modify attributes through an object instance.

```
Main myObj = new Main();  
System.out.println(myObj.x); // Accessing Value  
myObj.x = 40;                // Modifying Value
```

The 'final' Keyword

If you wish to prevent an attribute's value from being modified (making it a constant), use the `final` keyword.

```
final tp int x = 10;  
myObj.x = 25; // ERROR: cannot assign a value to final variable
```

State Persistence

Each object maintains its own unique set of attributes. Changing an attribute in *Object A* does not affect *Object B*.

```
Main car1 = new Main();  
Main car2 = new Main();  
car2.x = 50; // car1.x remains 5
```

Quick Recap

- Attributes define the **State** of an object.
- Dot notation is required for access: `object.attribute`.
- Use `final` for read-only fields.